NAME : .....

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)
RACE Regular Analysis through Continuous Exercise
JEE (Main + Advanced) 2024
ENTHUSIAST + LEADER COURSE

SCORE ADVANCED	MATHEMATICS	M.M. 102
REVISION ASSINGMENT # 02	(VECTOR-3D-02)	TIME : 90 MIN.

SECTION–I(i)

Straight Objective Type (3 Marks each, –1 for wrong answer)

1. Let $\vec{x}$, $\vec{y}$ are non zero and non collinear vectors such that

$(a\alpha^2 + b\alpha + c)\vec{x} + (a\beta^2 + b\beta + c)\vec{y} + (a\gamma^2 + b\gamma + c)(\vec{x} \times \vec{y}) = 0$, then $(a + b + c)$ is (where $a,b,c \in R$)

(A) –2 (B) –1 (C) 0 (D) 4

2. If for unit vectors $\hat{a}, \hat{b}$ and non-zero $\vec{c}$, $\hat{a} \times \hat{b} + \hat{a} = \vec{c}$ and $\hat{b}.\vec{c} = 0$, then volume of parallelopiped with coterminous edges $\hat{a}, \hat{b}$ and $\vec{c}$ will be (in cu.units)-

(A) 6 (B) 4 (C) 1 (D) $\frac{1}{2}$

3. If $\vec{d} = \lambda(\vec{a} \times \vec{b}) + \mu(\vec{b} \times \vec{c}) + \nu(\vec{c} \times \vec{a})$ $[\vec{a} \quad \vec{b} \quad \vec{c}] = \frac{1}{8}$ and $\vec{d}.(\vec{a} + \vec{b} + \vec{c}) = 8$, then $\lambda + \mu + \nu$ is

(A) 32 (B) 64 (C) 27 (D) 19

4. Let $\vec{V} = 2\hat{i} + \hat{j} - \hat{k}$, $\vec{W} = \hat{i} + 3\hat{k}$, $|\vec{U}| = 2$. If $\vec{U}$ is a vector in x-y plane, then greatest value of $([\vec{U} \ \vec{V} \ \vec{W}])^2$ is-

(A) 232 (B) 340 (C) 236 (D) 312

5. In a tetrahedron OABC, the edges are of lengths, $|OA| = |BC| = a$, $|OB| = |AC| = b$, $|OC| = |AB| = c$. Let G_1 and G_2 be the centroids of the triangle ABC and AOC such that $OG_1 \perp BG_2$, then the value of $\frac{a^2 + c^2}{b^2}$ is (O being origin)

(A) 2 (B) 3 (C) 6 (D) 9

6. The x-intercept of the plane which is passing through the intersection of the planes $x + 2y + 3z + 5 = 0$ and $2x - 3y + 7z + 1 = 0$ and parallel to line $\vec{r} = \hat{i} + 2\hat{j} + \lambda(8\hat{i} - 7\hat{j} - 4\hat{k})$, is-

(A) $\frac{13}{5}$ (B) $-\frac{7}{5}$ (C) $\frac{6}{5}$ (D) $-\frac{9}{5}$

7. The direction ratios of a normal to the plane passing through (1,0,0), (0,1,0) and making an angle $\frac{\pi}{4}$ with plane $x + y = 3$ can be-

(A) 0,1,0 (B) $1,1,\sqrt{2}$ (C) 1,0,0 (D) $\sqrt{2},1,1$

8. The equation of plane which passes thorugh the point of intersection of lines $\frac{x-1}{3} = \frac{y-2}{1} = \frac{z-3}{2}$ and $\frac{x-3}{1} = \frac{y-1}{2} = \frac{z-2}{3}$ and at greatest distance from point (0,0,0) is

(A) $4x + 3y + 5z = 25$ (B) $4x + 3y + 5z = 50$ (C) $3x + 4y + 5z = 49$ (D) $x + 7y - 5z = 2$

9. The direction cosines of the projection of the line $\frac{1}{2}(x-1) = -y = z + 2$ on the plane $2x + y - 3z = 4$ are-

(A) $\left(\frac{2}{\sqrt{6}}, -\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}\right)$ (B) $\left(\frac{2}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}\right)$ (C) $\left(\frac{2}{\sqrt{6}}, \frac{1}{\sqrt{6}}, -\frac{1}{\sqrt{6}}\right)$ (D) None of these

10. A unit vector $\hat{a}$ is such that $\hat{a} \times \vec{b} = \hat{i} - \hat{k}$, where $\vec{b}$ is $\hat{i} + \hat{j} + \hat{k}$, then the value of $|\hat{a}.\vec{b}|$ will be-

(A) 1 (B) 2 (C) 3 (D) 4

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SECTION–I(ii)

Multiple Correct Answer Type (4 Marks each, –1 for wrong answer)

11. If $\vec{a}=\hat{i}+6\hat{j}+3\hat{k}$; $\vec{b}=3\hat{i}+2\hat{j}+\hat{k}$ and $\vec{c}=(\alpha+1)\hat{i}+(\beta-1)\hat{j}+\hat{k}$ are linearly dependent vectors & $|\vec{c}|=\sqrt{6}$, then possible value(s) of $(\alpha+\beta)$ can be-

(A) 1 (B) 2 (C) 3 (D) 4

12. Let $a_1,a_2,a_3 \in R-\{0\}$ and $a_1+a_2\cos 2x+a_3\sin^2 x=0\ \forall\ x\in R$, then-

(A) vectors $\vec{a}=a_1\hat{i}+a_2\hat{j}+a_3\hat{k}$ and $\vec{b}=4\hat{i}+2\hat{j}+\hat{k}$ are perpendicular

(B) vectors $\vec{a}=a_1\hat{i}+a_2\hat{j}+a_3\hat{k}$ and $\vec{b}=\hat{i}+\hat{j}+2\hat{k}$ are parallel to each other

(C) if vector $\vec{a}=a_1\hat{i}+a_2\hat{j}+a_3\hat{k}$ is of length $\sqrt{6}$, then one of ordered triplet $(a_1,a_2,a_3)=(1,-1,-2)$

(D) if $2a_1+3a_2+6a_3=26$, then $|a_1\hat{i}+a_2\hat{j}+a_3\hat{k}|=2\sqrt{6}$

13. ΔABC is obtuse angled at B and points D and E are taken on sides AC and BC respectively such that AD : DC = 2 : 3 and BE : EC = 1 : 2. If DE produced meet AB produced at F, then -

(A) AF : BF = 4 : 3 (B) AF : BF = 3 : 1 (C) DE : EF = 1 : 5 (D) DE : EF = 3 : 4

14. If angle between the vectors $\vec{u}=x\hat{i}-3\hat{j}-\hat{k}$ and $\vec{v}=2x\hat{i}+x\hat{j}-\hat{k}$ is acute and the angle between the vector $\vec{v}$ and y-axis is obtuse, then x^2 may take the value -

(A) $\frac{1}{9}$ (B) 1 (C) 2 (D) $\sqrt[5]{3}$

15. Let $\bar{a}\times\bar{b}+\bar{b}=\bar{c}\times\bar{a}+\bar{b}\times\bar{c}$ where $|\bar{a}|=|\bar{b}|$ then identify the correct statement(s)

(A) $\bar{a}.\bar{b}=0$ (B) $\bar{b}.\bar{c}=0$ (C) $\bar{c}.\bar{a}=0$ (D) $\bar{c}=\bar{0}$

16. From point A(–1, 1, –1), the perpendicular is drawn to a plane P_1 whose foot is (1, –1, 1). From point B(3, –1, –1), the perpendicular is drawn to a plane P_2 whose foot is (1, 2, 1). P_3 is a plane containing the line of intersection of P_1 and P_2 and passing through the mid point of the line joining A and B. Then which of the following statement(S) is/are correct?

(A) Direction ratios of line of intersection of P_1 and P_2 are 5, 4, –1.

(B) Equation of plane P_3 is $16x - 19y + 4z = 12$

(C) Equation of line of intersection of P_1 and P_2 is $\frac{x-15}{5}=\frac{y-12}{4}=\frac{z-0}{-1}$

(D) The length of perpendicular from (0, 0, 0) to plane P_3 is $\frac{2}{\sqrt{18}}$

SECTION–I(iii)

Linked Comprehension Type (Single Correct Answer Type) (3 Marks each, –1 for wrong answer)

Paragraph for Question 17 and 18

Given $\vec{a}=\hat{i}+2\hat{j}-3\hat{k}$, $\vec{b}=2\hat{i}-\hat{j}+\hat{k}$ and L be a line whose equation is $\vec{r}\times\vec{b}=\vec{a}\times\vec{b}$.

17. The shortest distance of the point P(0,1,2) from the line L is equal to -

(A) $\frac{\sqrt{119}}{3}$ (B) 2 (C) $\frac{\sqrt{219}}{3}$ (D) $\frac{\sqrt{13}}{3}$

18. If 'C' be the curve in the x-y plane satisfying $\vec{R}.(2\hat{i}-2\hat{j}-\vec{R})=1$ (where $\vec{R}$ be the position vector of a variable point on the curve), then the minimum distance of line 'L' from the centre of curve 'C' is-

(A) $2\sqrt{3}-1$ (B) $2\sqrt{3}$ (C) 4 (D) $\sqrt{11}-1$

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Paragraph for Question 19 & 20

Consider a tetrahedron PQRS with PQ = 4 units. A plane 'P_1' is drawn perpendicular to PQ. Let P',R' & S' denotes the respective foot of perpendiculars drawn from P,R & S to the plane 'P_1' such that area of $\Delta P'R'S' = 15$ sq. units.

19. Volume of the tetrahedron PQRS (in cu. units) is -

(A) $\frac{47}{2}$ (B) 20 (C) 25 (D) $\frac{57}{2}$

20. If the distance of S' from plane PQR is 5 units then the distance of R from line through PQ is -

(A) 5 units (B) 6 units (C) 8 units (D) 10 units

Paragraph for Questions 21 and 22

Consider a plane P : $x + y + z = 3$ and a circle $x^2 + y^2 = 1$, $z = 0$

21. If $A(x_1, y_1, 0)$ and $B(x_2, y_2, 0)$ are the coordinates of the points on the circle which are at maximum and minimum distance from the plane P, then $\left|(x_1 + y_1) - (x_2 + y_2)\right|$ is equal to

(A) 0 (B) $\sqrt{2}$ (C) $2\sqrt{2}$ (D) 2

22. Image of the points on the circle in the plane P lies on

(A) $x + y - 2z = 1$ (B) $x + y + z = 5$ (C) $2x + y + 2z = 10$ (D) $2x + 2y - z = 6$

SECTION–I(v)

Matching list type (4 × 4) (Single option correct) (3 Marks each, –1 for wrong answer)

23. Match List-I with List-II and select the correct answer using the code given below the list.

	List-I		**List-II**
(P)	$\vec{a} = 2\hat{i} + \hat{k}$, $\vec{b} = \hat{i} + \hat{j} + \hat{k}$, $\vec{c} = 4\hat{i} - 3\hat{j} + 7\hat{k}$. Given $\vec{x} \times \vec{b} = \vec{c} \times \vec{b}$ and $\vec{x}.\vec{a} = 0$, then $\vec{x} - \vec{c}$ is	(1)	$\frac{1}{7}\left(5\hat{i} + 17\hat{j}\right)$
(Q)	If $\vec{a} = \hat{i} + 2\hat{j}, \vec{b} = 2\hat{i} - \hat{k}, \vec{x} \times \vec{a} = \vec{b} \times \vec{a}, \vec{x} \times \vec{b} = \vec{a} \times \vec{b}$, then $\sqrt{14}\frac{\vec{x}}{\lvert\vec{x}\rvert}$ is	(2)	$\hat{i} - \frac{7}{5}\hat{j} + \frac{1}{5}\hat{k}$
(R)	A line passes through the points A and B whose position vectors are $\hat{i} - 2\hat{j} + \hat{k}$ and $\hat{i} + \hat{j} - 3\hat{k}$ respectively. The position vector of a point lying on the line segment AB at unit distance from point A is	(3)	$3\hat{i} + 2\hat{j} - \hat{k}$
(S)	The position vectors of two vertices of a triangle are $-\hat{i} + 3\hat{j}$ and $2\hat{i} + 5\hat{j}$ and orthocentre is $\hat{i} + 2\hat{j}$,then the third vertex is	(4)	$-5\left(\hat{i} + \hat{j} + \hat{k}\right)$

Codes :

	P	Q	R	S
(A)	4	3	2	1
(B)	4	3	1	2
(C)	3	4	2	1
(D)	3	4	1	2

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24.

	List-I	List-II
(P)	Let volume of a tetrahedron ABCD is $\frac{81}{2}$ cube unit and volume of parallelopiped whose three coterminous edges are line segments joining centroid of any face of tetrahedron with centroids of its other three faces is V cubic unit, then V is	(1) 9
(Q)	If image of the point (1, 0, 1) in the plane $x - y - z = 1$ is (a,b,c) then $3(a - b + c)$ is equal to	(2) 8
(R)	Locus of all the points which are at a distance of 3 units from the line $\vec{r} = \lambda(\hat{i} + \hat{j} + \hat{k})$ is given by $x^2 + y^2 + z^2 - xy - yz - zx = \frac{k}{2}$, then find $\frac{k}{9}$	(3) 3
(S)	$\vec{a}, \vec{b}$ and $\vec{c}$ are such that $\lvert\vec{a}\rvert = \sqrt{3}, \lvert\vec{b}\rvert = 2$ and $\lvert\vec{c}\rvert = \sqrt{6}$. If $\vec{a}.\vec{b} < 0; \vec{c}$ is perpendicular to both $\vec{a}$ and $\vec{b}$ and $\vec{a}, \vec{b}, \vec{c}$ form the coterminous edges of a tetrahedron of unit volume, then angle between $\vec{a}$ and $\vec{b}$ is $\frac{p\pi}{q}$, (where p, q are coprime numbers), where (p + q) is	(4) 7

Code :

	P	Q	R	S
(A)	1	2	3	4
(B)	4	3	2	1
(C)	1	4	3	2
(D)	1	2	4	3

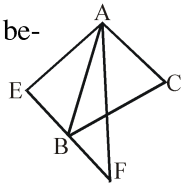
SECTION–II (iii)

Numerical Grid Type (Upto Second Decimal place) (4 Marks each, –1 for wrong answer)

1. Let $\vec{r} = \sin x(\vec{a} \times \vec{b}) + \cos y(\vec{b} \times \vec{c}) + 2(\vec{c} \times \vec{a})$, where $\vec{a}, \vec{b}$ & $\vec{c}$ are non-zero and non-coplanar vectors. If $\vec{r}$ is orthogonal to $\vec{a} + \vec{b} + \vec{c}$, then minimum value of $\frac{(x^2 + y^2)}{\pi^2}$ is

2. A unit vector $\hat{a}$ is such that $\hat{a} \times \vec{b} = \hat{i} - \hat{k}$, where $\vec{b}$ is $\hat{i} + \hat{j} + \hat{k}$. Value of $\lvert\hat{a}.\vec{b}\rvert$ will be-

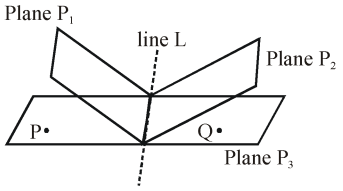
3. If ΔABC and ΔAEF are such that EB = BF, AB = EF = 1, BC = 6, $AC = \sqrt{33}$ and $\overrightarrow{AB}.\overrightarrow{AE} + \overrightarrow{AC}.\overrightarrow{AF} = 2$, then the value of $\overrightarrow{BC}.\overrightarrow{BF}$ is (given figure)



4. A plane passes point P(0, 1, –1) and contains a line $L: \frac{x-1}{2} = \frac{y-1}{3} = \frac{z+2}{1}$. If R is a point on line L such that ΔPQR is right angled at P, where Q(1, 1, –2), then the value of $\frac{PR^2}{5}$ is

5. In a tetrahedron LMNO edges ML, MN & MO are mutually perpendicular and length of altitudes drawn from O, L and N to opposite faces are 1,2 and 3 units respectively, then length of altitude drawn from M to face LNO is -

6. In adjacent diagram equation of plane P_1 and P_2 are $x + y - 2z = 1$ and $\alpha x - y + z = 2$ respectively (where $\alpha \in Q$). Coordinates of points P and Q are $(\alpha,1,3)$ and $\left(\frac{5}{3},1,\alpha\right)$ respectively on the plane P_3. if ℓ,m,n are direction cosines of line L then absolute value of $\frac{\sqrt{35}(\ell + m + n)}{\alpha}$ is



REVISION ASSINGMENT # 01 (VECTOR-3D-01) MATHEMATICS

	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-I	A.	A	D	D	C	D	B,C	B,D	A,B,C	A,C	A,D
	Q.	11	12	13							
	A.	A	C	C							
SECTION-II	Q.	1	2	3	4	5					
	A.	3.96	3.00	0.5	1.00	3.12 or 3.13					
SECTION-IV	Q.1	A	B	C	D						
		Q,S	T	P	R						

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